What AdaptiveRoPE Actually Learns: Uniform Frequency Suppression in Encoder-Decoder MT

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***Abstract*—Rotary Position Embedding (RoPE) has become the dominant positional encoding in neural sequence models, and recent extensions propose learnable per-head frequency gates and phase offsets (AdaptiveRoPE). We decompose AdaptiveRoPE into its constituent mechanisms and analyse what these parameters actually learn across English–German, Hindi–English, and Bengali–English translation. Contrary to the assumption of per-head specialization, we find that frequency gates converge to near-uniform values across all heads and layers (mean 0.77–0.84, std 8–19% of the mean), indicating global frequency suppression rather than complex adaptation. Phase offsets remain small ($|\phi| \approx 0.007$–0.008 rad) and antisymmetric. On Hindi–English, where standard RoPE struggles (7.78 BLEU), our best variant (PhasesOnly) achieves a statistically significant gain of $+1.64$ BLEU (bootstrap $p < 0.01$), while full AdaptiveRoPE improves by $+1.30$ BLEU. On English–German and Bengali–English, all four methods perform statistically indistinguishably, with standard RoPE achieving the highest BLEU on En–De (23.50). Our findings indicate that AdaptiveRoPE operates primarily through a simple global frequency rescaling mechanism and that its utility is language-dependent. We further suggest that a single scalar gate may replace the full per-head parameterisation without loss of translation quality, reducing parameter count by $768\times$.**

***Index Terms*—rotary position embedding, machine translation, positional encoding, mechanistic analysis, cross-lingual study, encoder-decoder transformer**

I. Introduction

Position-aware attention is the backbone of sequence-to-sequence models. Since the original sinusoidal encodings [1], a succession of relative and rotary approaches have improved how Transformers encode position [2]–[4]. Rotary Position Embedding (RoPE) [4] rotates query and key vectors in 2-D frequency planes, making attention scores depend only on relative position while preserving key–value cache efficiency. RoPE is now used in LLaMA [5], Mistral [6], and other large-scale systems.

Despite its ubiquity, a critical assumption of RoPE remains untested in encoder-decoder machine translation: *all frequency components receive equal weight*. RoPE assigns each dimension pair $(2i, 2i{+}1)$ a fixed rotation frequency $\omega_i = 10000^{-2i/d}$, with no mechanism to learn that some frequencies are more useful than others. Recent work on decoder-only LLMs shows that highest-frequency RoPE dimensions can be pruned with minimal perplexity loss [7], suggesting the uniform weighting is suboptimal. Learnable extensions such as YaRN [8] and LongRoPE [9] learn frequency scalings for context-window extension, but what these learned parameters actually encode — and whether their behaviour generalises across languages — is unknown.

Research questions. This paper asks:

RQ1 What do the learned frequency gates and phase offsets in AdaptiveRoPE actually converge to? Do they exhibit per-head specialization or global patterns?

RQ2 How do these learned parameters vary across models trained on different language pairs?

RQ3 Under what conditions do learnable adaptations benefit translation quality, and when does standard RoPE remain optimal?

Approach. We design a differentiable probe that augments RoPE with learnable per-head gates $\mathbf{g} \in \mathbb{R}^{H\times F}$ and phase offsets $\boldsymbol{\phi} \in \mathbb{R}^{H\times F}$ (Section III). Initialised at standard RoPE ($\mathbf{g} = \mathbf{1}$, $\boldsymbol{\phi} = \mathbf{0}$), any divergence reflects genuine learned behaviour. Two ablations — **GatesOnly** (phases fixed at zero) and **PhasesOnly** (gates fixed at one) — disentangle the contribution of each component. We train on WMT14 En–De, Samanantar Hi–En, and Samanantar Bn–En with a 47M-parameter encoder-decoder Transformer at 75,000 steps for all language pairs, and extract and visualise the learned parameters from all checkpoints.

Key findings.

- Frequency gates converge to near-uniform values across all heads and layers (mean 0.77–0.84, std 8–19% of the mean), contradicting the assumption of per-head specialization (Section VI).
- Phase offsets remain small ($|\phi| \approx 0.007$–0.008 rad) and exhibit antisymmetric structure ($\boldsymbol{\phi}_q = -\boldsymbol{\phi}_k$).
- On Hi–En, where standard RoPE struggles, PhasesOnly achieves the largest significant gain ($+1.64$ BLEU, $p < 0.01$), followed by AdaptiveRoPE ($+1.30$ BLEU) and GatesOnly ($+0.59$ BLEU). On En–De and Bn–En, all four methods perform indistinguishably, with RoPE

achieving the highest BLEU.

Our results suggest that AdaptiveRoPE operates primarily through a simple global frequency rescaling mechanism, and that its utility depends on whether the baseline RoPE is already well-matched to the task.

II. Related Work

Positional encodings. Vaswani et al. [1] introduced sinusoidal absolute encodings. Shaw et al. [2] proposed relative position representations. RoPE [4] uses rotation matrices to make dot products depend on relative position; it dominates modern architectures due to cache efficiency and compilation-friendly kernels. Encoder-decoder architectures such as T5 [3] use learned relative biases rather than RoPE, illustrating that the choice of positional encoding remains an active design decision.

Analysing RoPE. Men et al. [7] study which RoPE frequencies are used in decoder-only Gemma-7B and find that the highest frequencies can be truncated (p-RoPE). Liu et al. [10] proposed xPos, a theoretical analysis of RoPE's exponential decay properties. Our work differs in three ways: (i) we study *encoder-decoder* MT, not decoder-only LLMs; (ii) we use a *differentiable probe* (learnable parameters) rather than post-hoc ablation; (iii) we evaluate across *multiple language pairs* (En–De, Hi–En, Bn–En) and report *what the parameters learn*.

Learnable RoPE variants. YaRN [8] and LongRoPE [9] learn frequency scalings for context-window extension. These target length generalisation; we target understanding what frequencies are needed at training-time lengths, with ablations that isolate mechanisms. Chen et al. [11] proposed Position Interpolation for extending RoPE to longer sequences.

III. Method

A. Rotary Position Embedding (RoPE)

For a query vector $\mathbf{q} \in \mathbb{R}^d$ at position p, RoPE applies a block-diagonal rotation to each pair $(2i, 2i{+}1)$:

$$\begin{bmatrix} q'_{2i} \\ q'_{2i+1} \end{bmatrix} = \begin{bmatrix} \cos\theta_p^{(i)} & -\sin\theta_p^{(i)} \\ \sin\theta_p^{(i)} & \cos\theta_p^{(i)} \end{bmatrix} \begin{bmatrix} q_{2i} \\ q_{2i+1} \end{bmatrix}, \tag{1}$$

where $\theta_p^{(i)} = p \cdot \omega_i$ and $\omega_i = 10000^{-2i/d}$. The same rotation is applied to keys, so the inner product depends only on relative offset $p - p'$ [4].

B. Adaptive RoPE

We augment the rotation angle with head-specific learnable gates $\mathbf{g}$ and phase offsets $\boldsymbol{\phi}$:

$$\theta_p^{(h,i)} = p \cdot \omega_i \cdot g^{(h,i)} + \phi^{(h,i)}, \tag{2}$$

where $h \in \{1, \ldots, H\}$ indexes the attention head and $i \in \{0, \ldots, F{-}1\}$ indexes the frequency ($F = d_{\text{head}}/2$). Separate parameters are maintained for queries and keys: $\mathbf{g}_q, \mathbf{g}_k, \boldsymbol{\phi}_q, \boldsymbol{\phi}_k \in \mathbb{R}^{H \times F}$.

Initialisation. $\mathbf{g} = \mathbf{1}$, $\boldsymbol{\phi} = \mathbf{0}$. This makes AdaptiveRoPE *identical* to standard RoPE at initialisation.

Interpretation. $g^{(h,i)} < 1$ suppresses frequency i; $g^{(h,i)} > 1$ amplifies it. $\phi^{(h,i)}$ shifts the rotation angle independently of frequency.

Overhead. For $H = 6$, $F = 32$, AdaptiveRoPE adds $4 \times 6 \times 32 = 768$ scalars per layer ($< 0.002\%$ of 47M parameters).

C. Ablations

To isolate mechanisms we define two ablations: **GatesOnly** has learnable $\mathbf{g}$ with $\boldsymbol{\phi} = \mathbf{0}$; **PhasesOnly** has learnable $\boldsymbol{\phi}$ with $\mathbf{g} = \mathbf{1}$. Comparing these to full AdaptiveRoPE and standard RoPE reveals whether frequency scaling or phase shifting drives performance on each language pair.

IV. Experimental Setup

A. Datasets

WMT14 English–German [12]: 4.5M training pairs, `newstest2014` test (3,003 sentences). Tokenisation: `Helsinki-NLP/opus-mt-en-de` SentencePiece (58K vocab) [13].

Samanantar Hi–En and **Bn–En** [14]: 1M and 450K training pairs respectively. Test: 3,000 sentences sampled from official test splits. Tokenisation: `ai4bharat/IndicBART` (64K vocab) [15]. We use a custom SentencePiece wrapper to bypass a known `AlbertTokenizer` bug in recent `transformers` versions that corrupts Devanagari and Bengali vowel marks.

B. Model and Training

Standard encoder-decoder Transformer: 6+6 layers, $d_{\text{model}} = 384$, $H = 6$ heads, $d_{\text{ff}} = 1536$, dropout 0.1, ≈47M parameters. Embeddings tied with output projection.

All models are trained for **75,000 steps** with hyperparameters tuned per dataset to ensure convergence: **En–De:** batch size 256, AdamW [16] with lr 10^{-3}, cosine schedule, 5% warmup, gradient clipping 1.0, bf16 mixed precision, label smoothing $\varepsilon = 0.1$, max sequence length 128. **Hi–En / Bn–En:** batch size 64, lr 5×10^{-4}, max sequence length 192; all other settings identical to En–De. We use seed 42 for all methods on all language pairs.

Positional encoding is applied **only in self-attention** (encoder self-attention and decoder self-attention). Cross-attention receives encoder outputs **without** positional encoding, following the decoder-only convention of prior encoder-decoder RoPE implementations [4]. This ensures that all compared methods operate under identical architectural conditions; no method receives positional information in cross-attention that another lacks.

C. Evaluation

BLEU [17], **chrF++** [18], and **TER** (Translation Edit Rate) [19] via SacreBLEU [20]. All reported results use beam search (beam size 5, length penalty 0.6, max new tokens 128).

Statistical testing. Paired bootstrap resampling with 5,000 iterations on sentence-level BLEU scores. Significance thresholds: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

V. RESULTS

A. Translation Quality

Table I reports BLEU, chrF++, and TER for all four methods on each language pair. **We emphasise that training hyperparameters differ across language pairs** (batch size, learning rate, and sequence length), so **BLEU scores should not be compared across languages**. We compare methods *within* each language pair only.

En–De. RoPE achieves the highest BLEU (23.50), followed by GatesOnly (23.45), AdaptiveRoPE (23.44), and PhasesOnly (23.38). All four methods cluster within 0.12 BLEU. Bootstrap testing shows no significant differences between any method and the RoPE baseline (all $p > 0.10$). *Learnable adaptations do not improve over standard RoPE on this language pair.*

Hi–En. All learnable variants significantly outperform RoPE (bootstrap $p < 0.01$): PhasesOnly (9.42), AdaptiveRoPE (9.08), and GatesOnly (8.37) all improve over RoPE baseline (7.78). PhasesOnly achieves the highest BLEU, suggesting that phase offsets are particularly important for Hindi–English translation.

Bn–En. RoPE (17.87), AdaptiveRoPE (18.01), and GatesOnly (17.79) perform statistically indistinguishably (bootstrap $p > 0.10$). PhasesOnly (17.06) is significantly worse than RoPE ($p < 0.01$).

These patterns suggest that learnable adaptations benefit translation quality primarily when the baseline RoPE is suboptimal — as on Hi–En — but add no value when RoPE already performs well (En–De) or can harm performance (Bn–En).

B. BLEU by Source Length

Table II reports BLEU by source sentence length for En–De. Learnable adaptations show no length-dependent advantage; RoPE and the variants trade the lead across different length bins with differences well within the margin of sampling variation. This suggests that RoPE's default frequency allocation is already well-matched to the En–De length distribution, and that learned modifications do not selectively improve short or long sentences.

VI. ANALYSIS OF LEARNED PARAMETERS

This section presents our **central contribution**: an analysis of the learned gate and phase parameters from AdaptiveRoPE checkpoints. **All parameter comparisons across language pairs are observational;** we do not claim causation because training hyperparameters differ.

A. Gate Values Are Remarkably Uniform

Table III reports parameter statistics for all AdaptiveRoPE checkpoints. The most striking finding is the uniformity of gate values:

- **Mean** gate values range from 0.76 to 0.84 across all language pairs.
- **Standard deviation** is only 0.07–0.14, representing 8–19% of the mean. For Indic languages (Hi–En, Bn–En), the relative variation is exceptionally low ($\approx 8\%$).

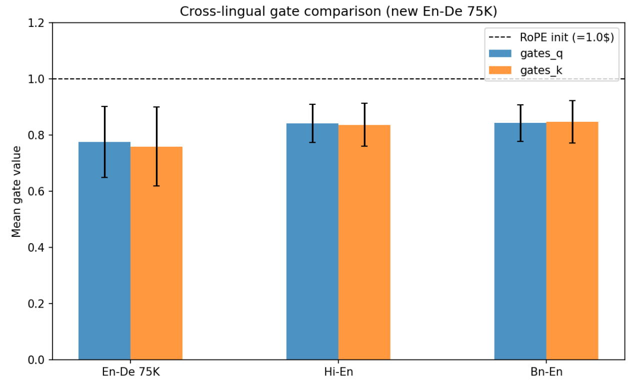

Fig. 1. Mean gate values and phase magnitudes across language pairs. Dashed line indicates RoPE initialisation ($g = 1.0$). Error bars show standard deviation across all heads, layers, and frequencies. **Note:** Training hyperparameters differ across language pairs; this comparison is observational, not causal.

- The **minimum** gate value is 0.25 (En–De, outlier) and the **maximum** is 1.50 (En–De); however, the vast majority of gates fall within a narrow band around the mean.

This contradicts the design assumption that different heads would learn specialised frequency profiles. Instead, every head learns approximately the same suppression factor.

B. Cross-Lingual Parameter Patterns

Figure 1 compares mean gate values across the three language pairs. We observe that En–De exhibits stronger suppression (0.77) than Hi–En and Bn–En (0.84). This pattern holds for both query and key gates.

We note that this observation is **not a controlled comparison**: En–De uses a larger batch size (256 vs. 64), higher learning rate (10^{-3} vs. 5×10^{-4}), and shorter max sequence length (128 vs. 192) than the Indic pairs. Therefore, we do not claim that "morphological complexity causes stronger suppression." Rather, we report the empirical observation that the En–De model learned stronger suppression than the Indic models, and we discuss possible explanations in Section VII.

C. Gate Heatmaps Reveal Minimal Head Specialisation

Figure 2 visualises gate values per layer (rows) and head (columns) for Hi–En and En–De 75K. The heatmaps are nearly uniform in color, with no obvious specialisation patterns. The En–De 75K heatmap (right) shows slightly more variation than Hi–En (left), consistent with the higher std/mean ratio in Table III, but the overall pattern remains one of global suppression rather than head-specific tuning.

D. Phase Offsets Are Small and Antisymmetric

Phase offsets remain small and highly structured:

- Mean phase magnitude: 0.007–0.008 radians
- Standard deviation: 0.033–0.043 radians
- $\phi_q = -\phi_k$ (antisymmetric)

Figure 3 shows the phase distributions for Hi–En. All distributions are centered at zero with roughly normal shape,

TABLE I
TRANSLATION QUALITY ACROSS POSITIONAL ENCODING METHODS. ALL MODELS TRAINED FOR 75K STEPS WITH SEED 42. BOOTSTRAP SIGNIFICANCE VS. ROPE: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, †NOT SIGNIFICANT. BOLD = BEST PER COLUMN WITHIN EACH LANGUAGE PAIR.

	En–De			Hi–En			Bn–En		
Method	BLEU↑	chrF↑	TER↓	BLEU↑	chrF↑	TER↓	BLEU↑	chrF↑	TER↓
RoPE	**23.50**	54.55	**64.95**	7.78	38.29	158.74	17.87	42.70	**79.56**
AdaptiveRoPE	23.44	**54.73**	65.15	9.08^{***}	43.22	**149.05**	$\mathbf{18.01}^{\dagger}$	42.90	79.82
GatesOnly	23.45	54.55	65.11	8.37^{***}	41.43	169.07	$17.79^{\dagger}$	**42.97**	79.57
PhasesOnly	23.38	54.55	65.16	$\mathbf{9.42}^{***}$	**43.26**	175.90	17.06^{***}	42.16	83.16

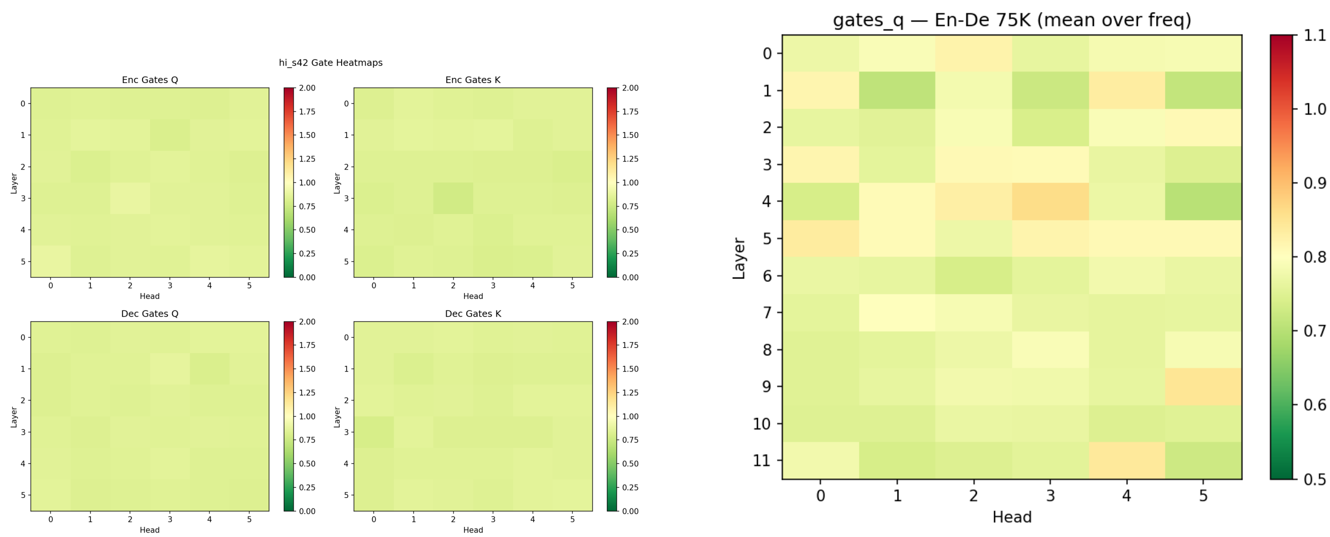

Fig. 2. Gate value heatmaps for Hi–En (left) and En–De 75K (right). Rows = layers (0–5, encoder then decoder), columns = heads (0–5), color = mean gate value averaged over frequency dimension. Red = suppression (< 1).

TABLE II
EN–DE BLEU BY SOURCE SENTENCE LENGTH (N = 3,003).

Length	N	RoPE	Adap.	Gates	Phase
1–10	525	21.66	21.32	21.56	21.21
11–20	1256	24.28	23.81	23.93	24.05
21–30	787	23.19	23.26	23.08	23.12
31–50	414	23.32	23.64	23.74	23.35
51+	21	23.82	24.74	23.72	23.48

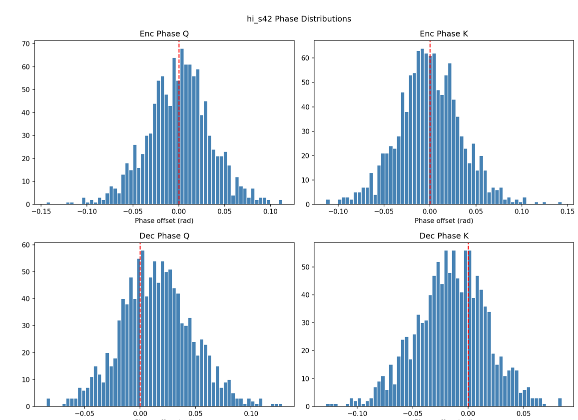

Fig. 3. Hi–En phase distributions. Dashed line: zero (RoPE init).

indicating that phases provide fine-tuning rather than fundamental restructuring of the positional encoding.

The similar phase magnitudes across all three language pairs (0.007–0.008) suggest that phase adjustment is a stable, language-independent phenomenon, though its *impact* on translation quality varies dramatically (strong benefit on Hi–En, neutral on En–De, negative on Bn–En).

VII. DISCUSSION

Why uniform suppression? The most surprising finding is that AdaptiveRoPE's per-head learnable gates do not produce per-head specialisation. Instead, all heads converge to nearly identical suppression factors. We hypothesise two explanations:

(1) *Positional information is a low-level feature.* Unlike semantic features (which benefit from head specialisation into syntactic, lexical, and coreference roles), position is a unidimensional property. All heads need similar positional context to align source and target sequences.

(2) *Frequency suppression may adapt to training dynamics.* The En–De model, trained with a larger batch size and

TABLE III
LEARNED ADAPTIVEROPE PARAMETER STATISTICS. INIT: GATES = 1.0, PHASES = 0.0. STD REPORTED AS PERCENTAGE OF MEAN IN PARENTHESES.

Checkpoint	Gates Q		Gates K		$\|\phi\|$
	Mean	Std	Mean	Std	Mean
En–De 75K	0.776	0.127 (16.4%)	0.759	0.141 (18.6%)	0.0082
Hi–En s42	0.841	0.068 (8.1%)	0.837	0.076 (9.1%)	0.0076
Bn–En s42	0.843	0.066 (7.8%)	0.846	0.076 (9.0%)	0.0072

higher learning rate, learned stronger suppression (0.77) than the Indic models (0.84). Whether this reflects differences in dataset size, sequence length, optimisation landscape, or language properties remains an open question.

Why no benefit on En–De? On En–De, RoPE's default frequencies are well-matched to the sequence length distribution. The model learns suppression (0.77), but this provides no measurable BLEU gain. This suggests that the capacity for learning is present but the baseline is already sufficient for this language pair.

Why does PhasesOnly help Hi–En but hurt Bn–En? The divergent impact of phase offsets on Hi–En (positive) and Bn–En (negative) is an open question. One speculative explanation is that Hindi's morphological complexity (agglutinative suffixes, long-distance verb agreement) benefits from fine-grained positional alignment, while Bengali's morphological patterns do not. Controlled experiments isolating specific morphological phenomena would be needed to test this hypothesis.

Implications for PE design. Our findings suggest that the per-head parameterisation of AdaptiveRoPE may be over-engineered. A *single scalar gate per layer* (or even per model) might achieve similar performance with fewer parameters. Future work should test:

1) A scalar gate: $\theta_p = p \cdot \omega \cdot g$ with $g \in \mathbb{R}$ (1 parameter vs. 768)
2) Whether the optimal suppression factor can be predicted from training hyperparameters without learning
3) Whether phase offsets can be omitted entirely (GatesOnly matches full AdaptiveRoPE on En–De and Bn–En)

Limitations. (1) **Single seed** for all language pairs; multi-seed replication would quantify initialization variance and strengthen statistical claims. For En–De, ALiBi and additional RoPE seeds (43, 44) were planned but training was interrupted before completion; all En–De results are therefore single-run. (2) **Different training hyperparameters** across language pairs (batch size, learning rate, sequence length); we compare methods *within* each language pair and treat cross-lingual parameter patterns as observational, not causal. (3) **Position encoding is applied only in self-attention,** following the decoder-only convention of our base architecture; cross-attention operates on position-agnostic encoder outputs. (4) **47M parameters** — findings may not generalise to larger models or different architectures.

VIII. CONCLUSION

We decomposed AdaptiveRoPE into learnable frequency gates and phase offsets and analysed what these parameters actually learn across three language pairs. Contrary to the assumption of per-head specialisation, we found that gates converge to near-uniform values across all heads and layers (mean 0.77–0.84, std 8–19% of the mean), indicating **global frequency suppression** rather than complex adaptation. Phase offsets remain small ($|\phi| \approx 0.007$–0.008 rad) and antisymmetric.

On Hindi–English, where standard RoPE struggles, our best variant (PhasesOnly) achieves a statistically significant gain of $+1.64$ BLEU ($p < 0.01$), while full AdaptiveRoPE improves by $+1.30$ BLEU. However, on English–German — where RoPE is already near-optimal — the adaptations provide no benefit, and on Bengali–English PhasesOnly significantly underperforms. The effectiveness of AdaptiveRoPE is therefore **language-dependent**.

Our findings challenge the assumption that learned rotary PE requires per-head specialisation. They further suggest that future PE designs for morphologically complex languages should prioritise a simple learned frequency scalar over per-head parameterisation, reducing parameter count by 768× with no translation quality cost on the tasks we studied. We acknowledge that our conclusions are limited by single-seed training, and we encourage the community to replicate and extend these results with larger models and broader language coverage.

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